

Jorge Luis Vargas Lara

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EDUCATION

University of California, Davis

Davis, CA

M.S. in Water Resources Engineering, Dec. 2025 — B.S. in Civil and Environmental Engineering, June 2024

EXPERIENCE

Intern I

Sept 2025 – Present

Davids Engineering

Davis, CA

- Analyzed surface water use, evapotranspiration, and hydrologic inputs for SGMA Annual Reports using Python, Excel, and QGIS to support regulatory reporting and water resources planning
- Investigated discrepancies in IWFM-related inputs and historical documentation, identifying data inconsistencies and preparing technical recommendations to improve reporting workflows and model reliability
- Developed data analysis workflows for projects including the Madera County Dry Well Program, forecasting call volumes and improving access to weather station data for internal planning and project support
- Conducted fieldwork including soil sampling and installation of remote sensors in agricultural wells to support monitoring of soil moisture, salinity, and crop field conditions

Graduate Researcher

July 2023 – Present

Morales Lab – CEE, University of California-Davis

Davis, CA

- Developed pore-scale computational fluid dynamics (CFD) models to investigate flow and contaminant transport in porous media systems
- Designed automated Python and Bash workflows to generate, run, and post-process large ensembles of simulations, improving reproducibility and reducing manual processing
- Analyzed and visualized > 100 GB of simulation and experimental data to quantify biofilm impacts on hydraulic conductivity and transport behavior
- Integrated high-resolution experimental imagery into 3D computational domains and communicated results through technical presentations and documentation

Roofing Assistant – Seasonal

June 2019 – Present

Valley Roofing

Yucaipa, CA

- Performed physically demanding fieldwork in high-temperature and elevated environments, maintaining productivity and safety under challenging conditions
- Followed strict safety protocols at heights, demonstrating situational awareness and risk management
- Collaborated with team members to complete residential roofing projects efficiently, often handling and transporting heavy materials across job sites
- Communicated directly with clients to provide updates and address questions, ensuring reliable service

PROJECTS

Groundwater Flow and Transport Model – Colton, CA | MODFLOW 6, FloPy, QGIS

May 2025

- Developed a groundwater flow and transport model using MODFLOW 6 and FloPy to simulate regional aquifer behavior under realistic boundary conditions
- Integrated geospatial datasets including monitoring wells, topography, and plume extents using QGIS
- Analyzed model outputs to evaluate groundwater flow dynamics and contaminant transport patterns

Multiphase Oxygen Transport in Porous Media | CFD, Python

Jan 2026

- Built on biofilm transport models to simulate oxygen diffusion in multiphase porous systems with gas bubbles
- Adapted existing Python workflows for image processing, domain generation, and transport analysis

Guest Lecturer – Introduction to Turbulence | Teaching, Documentation

Dec 2025

- Delivered a lecture on numerical fluid simulations, emphasizing model setup, validation, and interpretation
- Developed instructional materials and reusable simulation templates to support student learning

TECHNICAL SKILLS

Python, Bash — Flopy, OpenFOAM, QGIS — Git, Jupyter, Linux, Github — Pandas, NumPy, Matplotlib, Geopandas

AWARD

Johannes DeVries Graduate Student Award for Outstanding Student in Water Resources Engineering

KEY PUBLICATIONS

- Flow, Mixing, and Reaction Dynamics in Progressively Bioclogged Porous Media - Paper in preparation
- Sensitivity Analysis of Biofilm Permeability in Progressively Bioclogged Porous Media- Paper in preparation

LEADERSHIP

SCIENCE IS ART - CEE ART GALLERY

Founder and organizer of the first annual CEE Art Show, showcasing diverse research within the Civil and Environmental Engineering Department.

RELEVANT COURSEWORK

Turbulence - Water Resources Simulation - Soil Mechanics - Contaminant Fate and Transport in Porous Medium - Groundwater Systems Design - Sediment Transport - Hydrological Time Series Analysis - Deterministic Optimization and Design - Water Resource System Engineering- Jon Herman - Groundwater Modeling- Thomas Harter